Statistics/Data analysis

```
      name:  The_value_of_a_helping_hand_sup
       log:  C:\Users\hansg\Documents\working\SHARE data\weighting\output_sup_2.smcl
  log type:  smcl
 opened on:   5 Apr 2026, 13:16:58
C:\Users\hansg\Documents\working\SHARE data\weighting
(1 observation deleted)
(10,398 real changes made)
(6,148 real changes made)
(70,185 real changes made)
(8,277 real changes made)
(23,812 real changes made)
(1,763 real changes made)
(1,199 real changes made)
(1,763 real changes made)

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
         age |    109,546         0.0000         0.0000             .          .

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
       thinc |    109,546         0.0000         0.0000             .          .

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
        yedu |    109,546         0.0000         0.0000        104.09     0.0000
(11,207 real changes made)

Summary for variables: id
Group variable: htype (Household type)

           htype |         N
-----------------+----------
             1 S |     37717
          1 CNRP |      8553
           1 C2R |     62737
    1 CNRP + 1 S |        17
         Multi S |       372
    1 C2R  + 1 S |       117
  1 C2R + 1 CNRP |         2
       Multi C2R |        29
 Multi S + 1 C2R |         2
-----------------+----------
           Total |    109546
-----------------------------------------------------------------------------

  Household
       type |      Freq.     Percent        Cum.
------------+-----------------------------------
          1 |     37,717       34.43       34.43
          2 |      8,553        7.81       42.24
          3 |     62,737       57.27       99.51
          4 |         17        0.02       99.52
          5 |        372        0.34       99.86
          6 |        117        0.11       99.97
          7 |          2        0.00       99.97
          8 |         29        0.03      100.00
         10 |          2        0.00      100.00
------------+-----------------------------------
      Total |    109,546      100.00
(37,717 real changes made)
(62,737 real changes made)
```

```
note: 2.naly omitted because of collinearity.
note: 2.rhfo#0.ghto identifies no observations in the sample.
note: 2.rhfo#1.ghto identifies no observations in the sample.
note: 2.rhfo#2.ghto omitted because of collinearity.
Iteration 0:  Log likelihood = -155909.88
Iteration 1:  Log likelihood = -129411.95
Iteration 2:  Log likelihood = -127810.63
Iteration 3:  Log likelihood = -127800.78
Iteration 4:  Log likelihood = -127800.77

Ordered logistic regression                         Number of obs =  109,546
                                                    LR chi2(37)   = 56218.21
                                                    Prob > chi2   =   0.0000
Log likelihood = -127800.77                         Pseudo R2     =   0.1803
```

sphus	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]	
age	.0860289	.0136663	6.29	0.000	.0592434	.1128145
c.age#c.age	-.0003938	.0000905	-4.35	0.000	-.000571	-.0002165
yedu	-.0052566	.0049296	-1.07	0.286	-.0149185	.0044054
c.yedu#c.yedu	-.0015209	.0002159	-7.05	0.000	-.001944	-.0010978
thinc	-.1874806	.0138293	-13.56	0.000	-.2145857	-.1603756
c.thinc#c.thinc	.0243874	.00615	3.97	0.000	.0123337	.0364412
gender						
Female	-.0242519	.0126343	-1.92	0.055	-.0490147	.000511
1.covid	-.1191595	.0196664	-6.06	0.000	-.1577049	-.0806141
household						
Couple, both responding	.1727914	.0153004	11.29	0.000	.1428031	.2027797
Multiple, at least 1 responding	.1985161	.0242636	8.18	0.000	.1509603	.246072
cjs						
Not applicable	.5527522	.0586308	9.43	0.000	.4378379	.6676666
Retired	.3243795	.0295461	10.98	0.000	.2664702	.3822889
Unemployed	.2347881	.0877173	2.68	0.007	.0628654	.4067108
Permanently sick or disabled	1.470066	.0628747	23.38	0.000	1.346834	1.593298
Homemaker	.3860394	.0362325	10.65	0.000	.315025	.4570538
Other	.2478686	.0625499	3.96	0.000	.1252731	.370464
gali						
Limited	1.993831	.0139032	143.41	0.000	1.966581	2.021081
lifesat	-.3126707	.0189211	-16.52	0.000	-.3497554	-.275586
c.lifesat#c.lifesat	-.0001739	.0013178	-0.13	0.895	-.0027567	

```
>     .0024088
                                  |
                            naly  |
                     At least 1   |  -.3679433   .0190237   -19.34   0.000    -.4052291
>    -.3306575
                  Not applicable  |          0  (omitted)
                                  |
                            rhfo  |
                     At least 1   |   .4017225   .0184691    21.75   0.000     .3655238
>     .4379212
                  Not applicable  |  -.1676647   .0325373    -5.15   0.000    -.2314368
>    -.1038927
                                  |
                            ghto  |
                     At least 1   |  -.2379034   .0181803   -13.09   0.000    -.2735361
>    -.2022706
                  Not applicable  |    .023651   .0197832     1.20   0.232    -.0151234
>     .0624254
                                  |
                       rhfo#ghto  |
         At least 1#At least 1    |  -.1942653    .033478    -5.80   0.000     -.259881
>    -.1286497
      At least 1#Not applicable   |  -.0314511   .0461905    -0.68   0.496    -.1219829
>     .0590807
            Not applicable#None   |          0  (empty)
      Not applicable#At least 1   |          0  (empty)
  Not applicable#Not applicable   |          0  (omitted)
                                  |
                         country  |
                         Austria  |   .1858812   .0296699     6.26   0.000     .1277293
>     .2440331
                         Germany  |   .6502363   .0365364    17.80   0.000     .5786263
>     .7218463
                     Netherlands  |   .0720568   .0360069     2.00   0.045     .0014845
>     .1426291
                           Spain  |   .7703034   .0337944    22.79   0.000     .7040675
>     .8365392
                           Italy  |   .5735683    .032775    17.50   0.000     .5093305
>     .6378062
                          France  |   .4947341   .0295903    16.72   0.000     .4367381
>     .5527301
                         Denmark  |  -.2681629   .0348857    -7.69   0.000    -.3365377
>    -.1997881
                     Switzerland  |   .0623421   .0320135     1.95   0.051    -.0004031
>     .1250874
                         Belgium  |   .1070488   .0296303     3.61   0.000     .0489745
>      .165123
                  Czech Republic  |   .4240379   .0319373    13.28   0.000      .361442
>     .4866338
                        Slovenia  |   .5828942   .0349451    16.68   0.000     .5144031
>     .6513852
------------------------------------+----------------------------------------------------------------
  ------------
                           /cut1  |  -.6739021    .512471                      -1.678327
>     .3305226
                           /cut2  |   1.104422   .5125613                       .0998203
>     2.109024
                           /cut3  |   3.590217   .5128108                       2.585126
>     4.595308
                           /cut4  |   5.884567   .5127923                       4.879512
>     6.889621
------------------------------------+----------------------------------------------------------------
  ------------
```

Tests of proportional odds assumption

```
Number of obs            =  109,546
Number of predictors     =      37
Number of outcome levels =       5
Degrees of freedom       =     111
```

Test	chi2	P>chi2
Brant	**4066.38**	**0.000**
Likelihood-ratio	**4013.27**	**0.000**
Score	**4114.17**	**0.000**
Wald	**4133.87**	**0.000**
Wolfe-Gould	**4054.87**	**0.000**

(file **TestAssum_2.doc** not found)
Click to Open File: TestAssum_2.doc
note: **2.naly** omitted because of collinearity.
note: **2.rhfo#0.ghto** identifies no observations in the sample.
note: **2.rhfo#1.ghto** identifies no observations in the sample.
note: **2.rhfo#2.ghto** omitted because of collinearity.

Generalized Ordered Logit Estimates Number of obs = **109,528**
Wald chi2(1) = .
Prob > chi2 = .
Log pseudolikelihood = **-122288.91** Pseudo R2 = **0.1987**

(Std. err. adjusted for **6** cluste
> rs in **Qyear**)

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% con
Excellent					
age	**.1080138**	**.0892252**	**1.21**	**0.226**	**-.0668644**
>	**.2828919**				
c.age#c.age	**-.0005205**	**.0006292**	**-0.83**	**0.408**	**-.0017537**
>	**.0007128**				
yedu	**.0163216**	**.0224947**	**0.73**	**0.468**	**-.0277673**
>	**.0604105**				
c.yedu#c.yedu	**-.0015351**	**.0008189**	**-1.87**	**0.061**	**-.0031402**
>	**.00007**				
thinc	**-.0919921**	**.0357449**	**-2.57**	**0.010**	**-.1620508**
>	**-.0219334**				
c.thinc#c.thinc	**-.0115203**	**.0150029**	**-0.77**	**0.443**	**-.0409253**
>	**.0178848**				
gender					
Female	**.1370262**	**.0555976**	**2.46**	**0.014**	**.0280569**
>	**.2459955**				
covid					
1	**-.0022543**	**.0123714**	**-0.18**	**0.855**	**-.0265018**
>	**.0219933**				
household					
Couple, both responding	**.1570119**	**.0492438**	**3.19**	**0.001**	**.0604958**
>	**.2535281**				
Multiple, at least 1 responding	**.1720177**	**.0725527**	**2.37**	**0.018**	**.029817**
>	**.3142183**				
cjs					
Not applicable	**.289118**	**.5484398**	**0.53**	**0.598**	**-.7858044**
>	**1.36404**				
Retired	**.2814986**	**.1240268**	**2.27**	**0.023**	**.0384105**
>	**.5245867**				
Unemployed	**.0980618**	**.2681386**	**0.37**	**0.715**	**-.4274802**
>	**.6236038**				
Permanently sick or disabled	**.6165004**	**.1364395**	**4.52**	**0.000**	**.3490839**
>	**.883917**				
Homemaker	**.3804347**	**.2012428**	**1.89**	**0.059**	**-.013994**
>	**.7748634**				

```
                        Other |  -.2423315   .2957264    -0.82   0.413    -.8219445
>     .3372815
                              |
                         gali |
                      Limited |   2.029331   .0432794    46.89   0.000     1.944505
>     2.114157
                              |
                      lifesat |   .4922345   .0892054     5.52   0.000     .3173952
>     .6670739
                              |
            c.lifesat#c.lifesat |  -.0613382   .0057646   -10.64   0.000    -.0726367
>    -.0500397
                              |
                         naly |
                    At least 1 |  -.1601796   .0206954    -7.74   0.000    -.2007419
>    -.1196174
                              |
                         rhfo |
                    At least 1 |   .2037824   .0642243     3.17   0.002     .0779051
>     .3296597
                Not applicable |  -.1838401   .0367374    -5.00   0.000    -.2558442
>     -.111836
                              |
                         ghto |
                    At least 1 |  -.2018759   .0921263    -2.19   0.028    -.3824402
>    -.0213116
                Not applicable |  -.0550423   .0293079    -1.88   0.060    -.1124847
>     .0024001
                              |
                     rhfo#ghto |
        At least 1#At least 1 |  -.2056273    .082458    -2.49   0.013    -.3672421
>    -.0440125
     At least 1#Not applicable |  -.2168848   .0835828    -2.59   0.009    -.3807041
>    -.0530654
                              |
                      country |
                      Austria |   .7131688   .0868086     8.22   0.000     .5430271
>     .8833106
                      Germany |   1.373615   .1157122    11.87   0.000     1.146823
>     1.600407
                  Netherlands |  -.1299713   .0956581    -1.36   0.174    -.3174578
>     .0575152
                        Spain |   1.522081    .060889    25.00   0.000     1.402741
>     1.641422
                        Italy |   .6474164   .0808684     8.01   0.000     .4889172
>     .8059155
                       France |   .6775758   .0411104    16.48   0.000     .5970009
>     .7581508
                      Denmark |   .0474479   .0613035     0.77   0.439    -.0727048
>     .1676006
                  Switzerland |   .8353509   .0955405     8.74   0.000      .648095
>     1.022607
                      Belgium |   .6673445   .0525947    12.69   0.000     .5642607
>     .7704283
                Czech Republic |   1.426761   .1805581     7.90   0.000     1.072873
>     1.780648
                     Slovenia |   1.046595   .1950955     5.36   0.000     .6642152
>     1.428976
                              |
                        _cons |  -3.366134   3.479123    -0.97   0.333    -10.18509
>     3.452822
------------------------------+----------------------------------------------------------
  ____________
Very_good
                              |
                          age |     .03997    .033805     1.18   0.237    -.0262866
>     .1062267
                              |
                   c.age#c.age |  -.0000426   .0002201    -0.19   0.847     -.000474
>     .0003888
                              |
                         yedu |    .004679   .0146293     0.32   0.749    -.0239939
>     .0333519
```

```
                             |
                 c.yedu#c.yedu |  -.0022523   .0005806    -3.88   0.000    -.0033902
>    -.0011143
                             |
                         thinc |  -.1929693   .0592363    -3.26   0.001    -.3090703
>    -.0768684
                             |
             c.thinc#c.thinc |   .0007052   .0228078     0.03   0.975    -.0439972
>     .0454076
                             |
                       gender |
                       Female |   .0641175   .0231901     2.76   0.006     .0186659
>     .1095692
                             |
                        covid |
                            1 |   -.057093   .0272786    -2.09   0.036    -.1105581
>    -.0036279
                             |
                    household |
      Couple, both responding |   .2317054   .0269976     8.58   0.000     .1787911
>     .2846196
Multiple, at least 1 responding |    .276138   .0402681     6.86   0.000     .1972141
>      .355062
                             |
                          cjs |
              Not applicable |   .2214545   .3833131     0.58   0.563    -.5298254
>     .9727344
                     Retired |   .3398034   .0640946     5.30   0.000     .2141803
>     .4654265
                  Unemployed |   .1984048   .1308622     1.52   0.129    -.0580803
>       .45489
  Permanently sick or disabled |   .6441271   .1575473     4.09   0.000     .3353401
>      .952914
                   Homemaker |   .3046628   .0504772     6.04   0.000     .2057293
>     .4035963
                       Other |  -.1422362   .3336602    -0.43   0.670    -.7961981
>     .5117257
                             |
                        gali |
                     Limited |   1.833734    .029702    61.74   0.000     1.775519
>     1.891949
                             |
                      lifesat |   .0594451   .0942783     0.63   0.528     -.125337
>     .2442272
                             |
          c.lifesat#c.lifesat |  -.0274752    .006117    -4.49   0.000    -.0394643
>    -.0154862
                             |
                        naly |
                  At least 1 |   .0298323   .0431085     0.69   0.489    -.0546588
>     .1143233
                             |
                        rhfo |
                  At least 1 |   .1756515   .0470103     3.74   0.000     .0835131
>     .2677899
              Not applicable |  -.1657715   .0679022    -2.44   0.015    -.2988573
>    -.0326856
                             |
                        ghto |
                  At least 1 |  -.1469542   .0341953    -4.30   0.000    -.2139757
>    -.0799327
              Not applicable |  -.0148713   .0389573    -0.38   0.703    -.0912262
>     .0614836
                             |
                   rhfo#ghto |
      At least 1#At least 1 |  -.1011274    .147046    -0.69   0.492    -.3893323
>     .1870774
     At least 1#Not applicable |  -.1060294   .0563191    -1.88   0.060    -.2164129
>     .0043541
                             |
                     country |
                     Austria |   .2430139   .0432245     5.62   0.000     .1582955
```

>	.3277323					
	Germany	1.096062	.0665061	16.48	0.000	.9657122
>	1.226411					
	Netherlands	.3815255	.053996	7.07	0.000	.2756953
>	.4873558					
	Spain	1.043016	.0445064	23.44	0.000	.9557851
>	1.130247					
	Italy	.8925955	.046768	19.09	0.000	.8009319
>	.9842592					
	France	.8604681	.0324374	26.53	0.000	.796892
>	.9240442					
	Denmark	-.3747616	.0601908	-6.23	0.000	-.4927333
>	-.2567899					
	Switzerland	.3999412	.0401508	9.96	0.000	.3212472
>	.4786353					
	Belgium	.4962347	.0546215	9.08	0.000	.3891785
>	.6032908					
	Czech Republic	.9284355	.0693692	13.38	0.000	.7924745
>	1.064397					
	Slovenia	.9874771	.1197013	8.25	0.000	.752867
>	1.222087					
	_cons	-1.425561	1.236032	-1.15	0.249	-3.848139
>	.997017					
Good						
	age	.1423934	.0289251	4.92	0.000	.0857013
>	.1990856					
	c.age#c.age	-.0007966	.0001914	-4.16	0.000	-.0011717
>	-.0004215					
	yedu	-.0306228	.006803	-4.50	0.000	-.0439565
>	-.0172892					
	c.yedu#c.yedu	-.0005356	.0003464	-1.55	0.122	-.0012146
>	.0001434					
	thinc	-.2254099	.0310987	-7.25	0.000	-.2863622
>	-.1644575					
	c.thinc#c.thinc	.0461914	.0168238	2.75	0.006	.0132173
>	.0791656					
	gender					
	Female	-.0138281	.0186931	-0.74	0.459	-.0504658
>	.0228097					
	covid					
	1	-.1126925	.0228985	-4.92	0.000	-.1575726
>	-.0678123					
	household					
	Couple, both responding	.1507199	.0424599	3.55	0.000	.0675
>	.2339397					
	Multiple, at least 1 responding	.2679336	.0406624	6.59	0.000	.1882366
>	.3476305					
	cjs					
	Not applicable	.3903463	.2579525	1.51	0.130	-.1152312
>	.8959239					
	Retired	.264217	.1144646	2.31	0.021	.0398704
>	.4885635					
	Unemployed	.1213847	.1013549	1.20	0.231	-.0772671
>	.3200366					
	Permanently sick or disabled	1.042237	.0630781	16.52	0.000	.9186058
>	1.165867					
	Homemaker	.2867306	.1951438	1.47	0.142	-.0957442
>	.6692055					
	Other	.0123472	.252174	0.05	0.961	-.4819047
>	.5065992					

```
                                 |
                            gali |
                         Limited |   2.109511   .0156485   134.81   0.000      2.07884
>     2.140181
                                 |
                         lifesat |  -.2839725   .0586023    -4.85   0.000    -.3988308
>    -.1691141
                                 |
             c.lifesat#c.lifesat |  -.0013701   .0039444    -0.35   0.728     -.009101
>     .0063608
                                 |
                            naly |
                       At least 1 |   -.208963   .0528216    -3.96   0.000    -.3124914
>    -.1054346
                                 |
                            rhfo |
                       At least 1 |   .3995367   .0662614     6.03   0.000     .2696666
>     .5294067
                   Not applicable |  -.1539005   .0465906    -3.30   0.001    -.2452163
>    -.0625846
                                 |
                            ghto |
                       At least 1 |  -.2453227   .0529708    -4.63   0.000    -.3491435
>    -.1415018
                   Not applicable |  -.0097677   .0265445    -0.37   0.713    -.0617941
>     .0422586
                                 |
                        rhfo#ghto |
          At least 1#At least 1 |   -.130645   .1446671    -0.90   0.366    -.4141873
>     .1528973
      At least 1#Not applicable |      -.006   .0728475    -0.08   0.934    -.1487784
>     .1367785
                                 |
                         country |
                         Austria |  -.0367755    .099358    -0.37   0.711    -.2315135
>     .1579625
                         Germany |   .2910489   .0696923     4.18   0.000     .1544545
>     .4276433
                     Netherlands |  -.1192118   .0427138    -2.79   0.005    -.2029294
>    -.0354942
                           Spain |   .5574902   .0681341     8.18   0.000     .4239499
>     .6910305
                           Italy |   .5178936   .1138105     4.55   0.000     .2948291
>     .7409581
                          France |    .113553   .0486723     2.33   0.020      .018157
>      .208949
                         Denmark |  -.0794939   .0935232    -0.85   0.395     -.262796
>     .1038082
                     Switzerland |  -.3773624   .0884826    -4.26   0.000     -.550785
>    -.2039398
                         Belgium |  -.2827783   .0350809    -8.06   0.000    -.3515357
>     -.214021
                   Czech Republic |  -.1643852   .1288724    -1.28   0.202    -.4169705
>        .0882
                        Slovenia |   .1637881   .0815294     2.01   0.045     .0039934
>     .3235827
                                 |
                           _cons |  -5.432039   1.001017    -5.43   0.000    -7.393997
>    -3.470081
-------------------------------+----------------------------------------------------------------
Fair
                                 |
                             age |   .1726985   .0422556     4.09   0.000     .0898791
>     .2555179
                                 |
                       c.age#c.age |  -.0010167   .0002792    -3.64   0.000     -.001564
>    -.0004694
                                 |
                            yedu |  -.0222713   .0051694    -4.31   0.000    -.0324031
>    -.0121395
                                 |
                     c.yedu#c.yedu |  -.0007852   .0004309    -1.82   0.068    -.0016297
```

```
>     .0000593
                                  |
                           thinc  |  -.1669246   .0583342    -2.86   0.004    -.2812574
>    -.0525917
                                  |
                c.thinc#c.thinc   |   .0553146   .0427768     1.29   0.196    -.0285264
>     .1391556
                                  |
                          gender  |
                          Female  |  -.0771831   .0518825    -1.49   0.137     -.178871
>     .0245047
                                  |
                           covid  |
                               1  |   -.111946   .0445079    -2.52   0.012      -.19918
>    -.0247121
                                  |
                       household  |
        Couple, both responding   |    .124828   .0516955     2.41   0.016     .0235067
>     .2261493
Multiple, at least 1 responding   |   .0889433   .0558874     1.59   0.112    -.0205939
>     .1984806
                                  |
                             cjs  |
                Not applicable    |   .4432482    .173456     2.56   0.011     .1032806
>     .7832158
                        Retired   |   .2872474   .0786944     3.65   0.000     .1330092
>     .4414856
                      Unemployed  |  -.7290974    .142252    -5.13   0.000    -1.007906
>    -.4502886
   Permanently sick or disabled   |   1.455874   .0954282    15.26   0.000     1.268838
>      1.64291
                       Homemaker  |   .2371182   .0846191     2.80   0.005     .0712678
>     .4029686
                           Other  |   .1228547   .1423133     0.86   0.388    -.1560743
>     .4017837
                                  |
                            gali  |
                         Limited  |   3.000066   .0728101    41.20   0.000     2.857361
>     3.142772
                                  |
                          lifesat |  -.5439719   .0682697    -7.97   0.000     -.677778
>    -.4101658
                                  |
             c.lifesat#c.lifesat  |   .0186543   .0051623     3.61   0.000     .0085363
>     .0287723
                                  |
                             naly |
                       At least 1 |  -.5260293   .0900784    -5.84   0.000    -.7025797
>    -.3494788
                                  |
                             rhfo |
                       At least 1 |   .5664547   .0310599    18.24   0.000     .5055784
>      .627331
                Not applicable    |   -.169367   .1047792    -1.62   0.106    -.3747304
>     .0359964
                                  |
                             ghto |
                       At least 1 |  -.5862338   .0749079    -7.83   0.000    -.7330505
>    -.4394171
                Not applicable    |   .1477759   .0268124     5.51   0.000     .0952245
>     .2003273
                                  |
                        rhfo#ghto |
          At least 1#At least 1   |   .0164257   .1575208     0.10   0.917    -.2923094
>     .3251608
      At least 1#Not applicable   |    .075095    .037353     2.01   0.044     .0018845
>     .1483056
                                  |
                          country |
                          Austria |  -.1269873   .2119669    -0.60   0.549    -.5424348
>     .2884601
                          Germany |   .0432051   .1644598     0.26   0.793    -.2791302
```

```
>     .3655404
                  Netherlands | -.4460858   .1568751    -2.84   0.004    -.7535554
>    -.1386162
                        Spain |  .4584647   .1661865     2.76   0.006     .1327451
>     .7841842
                        Italy | -.0929375   .1541983    -0.60   0.547    -.3951605
>     .2092856
                       France |  .2300266   .1676837     1.37   0.170    -.0986274
>     .5586805
                      Denmark | -.0022902   .1956406    -0.01   0.991    -.3857387
>     .3811584
                  Switzerland | -.5022673   .1857638    -2.70   0.007    -.8663575
>     -.138177
                      Belgium | -.5026985   .0893333    -5.63   0.000    -.6777885
>    -.3276085
              Czech Republic | -.0314834   .2001575    -0.16   0.875    -.4237848
>      .360818
                     Slovenia |  .3373354   .1276193     2.64   0.008     .0872062
>     .5874646
                              |
                        _cons | -8.538756   1.664149    -5.13   0.000    -11.80043
>    -5.277084
------------------------------+----------------------------------------------------
  ___________

WARNING! 6 in-sample cases have an outcome with a predicted probability that is
less than 0. See the gologit2 help section on Warning Messages for more information.
results_imp1_go_2.xls
dir : seeout
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -2.627e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.529e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.144e+08  (not concave)
Iteration 3:  Log pseudolikelihood =  -92247313  (not concave)
Iteration 4:  Log pseudolikelihood =  -84941024  (not concave)
Iteration 5:  Log pseudolikelihood =  -82653307  (not concave)
Iteration 6:  Log pseudolikelihood =  -81315700  (not concave)
Iteration 7:  Log pseudolikelihood =  -80571013  (not concave)
Iteration 8:  Log pseudolikelihood =  -79997720  (not concave)
Iteration 9:  Log pseudolikelihood =  -79643392  (not concave)
Iteration 10: Log pseudolikelihood =  -79311145  (not concave)
Iteration 11: Log pseudolikelihood =  -78536097  (not concave)
Iteration 12: Log pseudolikelihood =  -78400197  (not concave)
Iteration 13: Log pseudolikelihood =  -78303263  (not concave)
Iteration 14: Log pseudolikelihood =  -78220690
Iteration 15: Log pseudolikelihood =  -77890078  (not concave)
Iteration 16: Log pseudolikelihood =  -77846975
Iteration 17: Log pseudolikelihood =  -77798565  (not concave)
Iteration 18: Log pseudolikelihood =  -77721917
Iteration 19: Log pseudolikelihood =  -77700519  (not concave)
Iteration 20: Log pseudolikelihood =  -77684064
Iteration 21: Log pseudolikelihood =  -77680977
Iteration 22: Log pseudolikelihood =  -77680973
Iteration 23: Log pseudolikelihood =  -77680973

Stereotype logistic regression                      Number of obs =  29,989
                                                    Wald chi2(35) = 2227.02
Log pseudolikelihood = -77680973                    Prob > chi2   =  0.0000
```

(1) **[phi1_1]_cons = 1**

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]
age	.2734403	.1290115	2.12	0.034	.0205824 .5262982
c.age#c.age	-.0014097	.0008842	-1.59	0.111	-.0031428 .0003234
yedu	-.01752	.0383051	-0.46	0.647	-.0925966 .0575566
c.yedu#c.yedu	-.0050001	.0017962	-2.78	0.005	-.0085207 -.0014796
thinc	-.4943742	.11224	-4.40	0.000	-.7143605 -.2743879
c.thinc#c.thinc	.0545952	.0518436	1.05	0.292	-.0470164 .1562069
gender					
Female	.0383162	.1077803	0.36	0.722	-.1729293 .2495617
0.covid	0	(omitted)			
household					
Couple, both responding	.3530441	.1425074	2.48	0.013	.0737347 .6323534
Multiple, at least 1 responding	.7275779	.194375	3.74	0.000	.3466098 1.108546
cjs					
Not applicable	-.1112536	.858109	-0.13	0.897	-1.793116 1.570609
Retired	.5995059	.2106314	2.85	0.004	.186676 1.012336
Unemployed	-.2200502	.4125801	-0.53	0.594	-1.028692 .588592
Permanently sick or disabled	3.06402	.4465231	6.86	0.000	2.18885 3.939189
Homemaker	.3969318	.2440337	1.63	0.104	-.0813655 .8752292
Other	-1.626455	.4729849	-3.44	0.001	-2.553489 -.6994219
gali					
Limited	5.396818	.1554359	34.72	0.000	5.09217 5.701467
lifesat	-.4765453	.1844958	-2.58	0.010	-.8381505 -.11494
c.lifesat#c.lifesat	-.0210544	.0130023	-1.62	0.105	-.0465384 .0044295
naly					
At least 1	-.9158723	.140669	-6.51	0.000	-1.191578 -.6401661
Not applicable	0	(omitted)			
rhfo					
At least 1	1.204172	.1763094	6.83	0.000	.8586116 1.549732
ghto					
At least 1	-.4985409	.1544135	-3.23	0.001	-.8011858 -.1958961

```
                 Not applicable |  -.0503337   .1242634    -0.41   0.685    -.2938854
>      .193218                  |
                                |
                      rhfo#ghto |
          At least 1#At least 1 |   -1.02884   .3357023    -3.06   0.002    -1.686805
>    -.3708761
      At least 1#Not applicable |  -.0019104    .289468    -0.01   0.995    -.5692572
>     .5654364                  |
                                |
                        country |
                        Austria |   -.285565   .1945583    -1.47   0.142    -.6668923
>     .0957623
                        Germany |   1.058437   .2218851     4.77   0.000     .6235501
>     1.493324
                    Netherlands |  -.1876946   .2087909    -0.90   0.369    -.5969173
>      .221528
                          Spain |   1.936438   .2253073     8.59   0.000     1.494843
>     2.378032
                          Italy |   .9994561   .2149663     4.65   0.000     .5781299
>     1.420782
                         France |   .9238605   .1939764     4.76   0.000     .5436737
>     1.304047
                        Denmark |  -1.336701   .2425337    -5.51   0.000    -1.812059
>     -.861344
                    Switzerland |  -.6022242   .2196985    -2.74   0.006    -1.032825
>     -.171623
                        Belgium |  -.0310947   .1944968    -0.16   0.873    -.4123016
>     .3501121
                 Czech Republic |   .7417804    .233489     3.18   0.001     .2841503
>      1.19941
                       Slovenia |   1.403798   .2796529     5.02   0.000     .8556886
>     1.951908
-------------------------------+----------------------------------------------------
  ____________
                        /phi1_1 |          1  (constrained)
                        /phi1_2 |   .8541515   .0178522    47.85   0.000     .8191619
>     .8891412
                        /phi1_3 |   .6286815   .0134129    46.87   0.000     .6023927
>     .6549703
                        /phi1_4 |   .3487972   .0119919    29.09   0.000     .3252935
>     .3723009
                        /phi1_5 |          0  (base outcome)
-------------------------------+----------------------------------------------------
  ____________
                        /theta1 |   9.719812   4.715436     2.06   0.039     .4777275
>      18.9619
                        /theta2 |   9.686084   4.033968     2.40   0.016     1.779652
>     17.59252
                        /theta3 |   8.952889   2.985886     3.00   0.003      3.10066
>     14.80512
                        /theta4 |   5.741763   1.673829     3.43   0.001     2.461118
>     9.022409
                        /theta5 |          0  (base outcome)
-------------------------------+----------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_11_2.xls
dir : seeout

Predictive margins                                      Number of obs = 29,989
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))
```

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]
_predict#rhfo#ghto					
1#None#None	.0526459	.0021136	24.91	0.000	.0485034 .0567884
1#None#At least 1	.0611936	.003025	20.23	0.000	.0552647 .0671225
1#None#Not applicable	.0534677	.0023888	22.38	0.000	.0487858 .0581496
1#At least 1#None	.0355998	.0024981	14.25	0.000	.0307037 .040496
1#At least 1#At least 1	.0580832	.0049841	11.65	0.000	.0483145 .0678519
1#At least 1#Not applicable	.0362398	.0031501	11.50	0.000	.0300658 .0424138
2#None#None	.1324474	.003408	38.86	0.000	.1257679 .1391269
2#None#At least 1	.1457578	.0047299	30.82	0.000	.1364873 .1550282
2#None#Not applicable	.1337717	.0038201	35.02	0.000	.1262844 .1412591
2#At least 1#None	.1023739	.0045627	22.44	0.000	.0934312 .1113166
2#At least 1#At least 1	.1410298	.007548	18.68	0.000	.126236 .1558235
2#At least 1#Not applicable	.1036068	.0058607	17.68	0.000	.09212 .1150935
3#None#None	.3888574	.0048878	79.56	0.000	.3792775 .3984373
3#None#At least 1	.3987866	.0054396	73.31	0.000	.3881251 .4094481
3#None#Not applicable	.3899448	.0051509	75.70	0.000	.3798491 .4000404
3#At least 1#None	.3576598	.0061509	58.15	0.000	.3456043 .3697153
3#At least 1#At least 1	.3955105	.0068266	57.94	0.000	.3821305 .4088904
3#At least 1#Not applicable	.3592043	.0078128	45.98	0.000	.3438916 .3745171
4#None#None	.3099734	.0046533	66.61	0.000	.3008531 .3190938
4#None#At least 1	.2952539	.0056913	51.88	0.000	.2840991 .3064087
4#None#Not applicable	.3085455	.0050427	61.19	0.000	.298662 .3184289
4#At least 1#None	.3396251	.0062412	54.42	0.000	.3273925 .3518576
4#At least 1#At least 1	.3005724	.0088076	34.13	0.000	.2833098 .317835
4#At least 1#Not applicable	.3385331	.0068496	49.42	0.000	.3251082 .351958
5#None#None	.1160759	.0036499	31.80	0.000	.1089223 .1232295
5#None#At least 1	.0990082	.0050192	19.73	0.000	.0891707 .1088456
5#None#Not applicable	.1142703	.0043403	26.33	0.000	.1057635 .1227771
5#At least 1#None	.1647414	.0071283	23.11	0.000	.1507702 .1787126
5#At least 1#At least 1	.1048041	.008986	11.66	0.000	.0871919 .1224163
5#At least 1#Not applicable	.162416	.0103132	15.75	0.000	.1422025 .1826295

```
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -2.527e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -2.120e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.366e+08  (not concave)
Iteration 3:  Log pseudolikelihood = -1.014e+08  (not concave)
Iteration 4:  Log pseudolikelihood =  -77570055  (not concave)
Iteration 5:  Log pseudolikelihood =  -68820104  (not concave)
Iteration 6:  Log pseudolikelihood =  -65121939  (not concave)
Iteration 7:  Log pseudolikelihood =  -63378578  (not concave)
Iteration 8:  Log pseudolikelihood =  -62380117  (not concave)
Iteration 9:  Log pseudolikelihood =  -61470176  (not concave)
Iteration 10: Log pseudolikelihood =  -60485618  (not concave)
Iteration 11: Log pseudolikelihood =  -60244068  (not concave)
Iteration 12: Log pseudolikelihood =  -59859722  (not concave)
Iteration 13: Log pseudolikelihood =  -59599821  (not concave)
Iteration 14: Log pseudolikelihood =  -59485435  (not concave)
Iteration 15: Log pseudolikelihood =  -59386622  (not concave)
Iteration 16: Log pseudolikelihood =  -59298404  (not concave)
Iteration 17: Log pseudolikelihood =  -59218998  (not concave)
Iteration 18: Log pseudolikelihood =  -59146985  (not concave)
Iteration 19: Log pseudolikelihood =  -59081174  (not concave)
Iteration 20: Log pseudolikelihood =  -59020330  (not concave)
Iteration 21: Log pseudolikelihood =  -58963934  (not concave)
Iteration 22: Log pseudolikelihood =  -58911076  (not concave)
Iteration 23: Log pseudolikelihood =  -58861447  (not concave)
Iteration 24: Log pseudolikelihood =  -58814397  (not concave)
Iteration 25: Log pseudolikelihood =  -58769761  (not concave)
Iteration 26: Log pseudolikelihood =  -58727055  (not concave)
Iteration 27: Log pseudolikelihood =  -58686195  (not concave)
Iteration 28: Log pseudolikelihood =  -58646819  (not concave)
Iteration 29: Log pseudolikelihood =  -58608890  (not concave)
Iteration 30: Log pseudolikelihood =  -58572135  (not concave)
Iteration 31: Log pseudolikelihood =  -58536543  (not concave)
Iteration 32: Log pseudolikelihood =  -58501906  (not concave)
Iteration 33: Log pseudolikelihood =  -58468230  (not concave)
Iteration 34: Log pseudolikelihood =  -58435354  (not concave)
Iteration 35: Log pseudolikelihood =  -58403293  (not concave)
Iteration 36: Log pseudolikelihood =  -58371922  (not concave)
Iteration 37: Log pseudolikelihood =  -58341262  (not concave)
Iteration 38: Log pseudolikelihood =  -58311216  (not concave)
Iteration 39: Log pseudolikelihood =  -58281804  (not concave)
Iteration 40: Log pseudolikelihood =  -58252953  (not concave)
Iteration 41: Log pseudolikelihood =  -58224682  (not concave)
Iteration 42: Log pseudolikelihood =  -58196936  (not concave)
Iteration 43: Log pseudolikelihood =  -58169731  (not concave)
Iteration 44: Log pseudolikelihood =  -58143026  (not concave)
Iteration 45: Log pseudolikelihood =  -58116837  (not concave)
Iteration 46: Log pseudolikelihood =  -58091131  (not concave)
Iteration 47: Log pseudolikelihood =  -58065923  (not concave)
Iteration 48: Log pseudolikelihood =  -58041187  (not concave)
Iteration 49: Log pseudolikelihood =  -58016936  (not concave)
Iteration 50: Log pseudolikelihood =  -57993152
Iteration 51: Log pseudolikelihood =  -57477620  (not concave)
Iteration 52: Log pseudolikelihood =  -57375171
Iteration 53: Log pseudolikelihood =  -57245585  (not concave)
Iteration 54: Log pseudolikelihood =  -57216074
Iteration 55: Log pseudolikelihood =  -57213231  (not concave)
Iteration 56: Log pseudolikelihood =  -57188411
Iteration 57: Log pseudolikelihood =  -57182928
Iteration 58: Log pseudolikelihood =  -57181980
Iteration 59: Log pseudolikelihood =  -57181965
Iteration 60: Log pseudolikelihood =  -57181965

Stereotype logistic regression                      Number of obs =   22,899
                                                    Wald chi2(35) =  1189.45
Log pseudolikelihood = -57181965                    Prob > chi2   =   0.0000
```

(1) **[phi1_1]_cons = 1**

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.4442969	.1577892	2.82	0.005	.1350357	.7535582
c.age#c.age	-.0026421	.0010585	-2.50	0.013	-.0047167	-.0005675
yedu	-.0922814	.043873	-2.10	0.035	-.1782709	-.0062919
c.yedu#c.yedu	-.0010999	.0020025	-0.55	0.583	-.0050248	.002825
thinc	-.6303764	.1310793	-4.81	0.000	-.8872871	-.3734657
c.thinc#c.thinc	.1619837	.06297	2.57	0.010	.0385649	.2854026
gender						
Female	.0440583	.1229504	0.36	0.720	-.19692	.2850366
0.covid	0	(omitted)				
household						
Couple, both responding	.4120942	.1554455	2.65	0.008	.1074265	.7167618
Multiple, at least 1 responding	.4309959	.2386553	1.81	0.071	-.0367599	.8987517
cjs						
Not applicable	1.705761	.7873299	2.17	0.030	.1626225	3.248899
Retired	.9351225	.2890496	3.24	0.001	.3685957	1.501649
Unemployed	.3703407	.6459578	0.57	0.566	-.8957133	1.636395
Permanently sick or disabled	3.372613	.5680399	5.94	0.000	2.259276	4.485951
Homemaker	1.097127	.336898	3.26	0.001	.4368188	1.757435
Other	.183322	.5298728	0.35	0.729	-.8552096	1.221854
gali						
Limited	5.054351	.1967421	25.69	0.000	4.668744	5.439958
lifesat	-.7064709	.1793286	-3.94	0.000	-1.057949	-.3549933
c.lifesat#c.lifesat	-.0013084	.0129099	-0.10	0.919	-.0266113	.0239945
naly						
At least 1	-.7316491	.1606605	-4.55	0.000	-1.046538	-.4167603
Not applicable	0	(omitted)				
rhfo						
At least 1	1.08419	.1969291	5.51	0.000	.6982158	1.470164
ghto						
At least 1	-.8134957	.1833063	-4.44	0.000	-1.172769	-.4542221

```
                 Not applicable |  -.1166128   .1413345    -0.83   0.409    -.3936233
>     .1603978
                                |
                      rhfo#ghto |
         At least 1#At least 1  |  -.6538464   .4092735    -1.60   0.110    -1.456008
>      .148315
      At least 1#Not applicable |   .0244196   .3507562     0.07   0.944      -.66305
>     .7118892
                                |
                        country |
                        Austria |   .7603244   .2156054     3.53   0.000     .3377455
>     1.182903
                        Germany |   2.050246    .254126     8.07   0.000     1.552169
>     2.548324
                    Netherlands |   .5552479    .225543     2.46   0.014     .1131918
>     .9973039
                          Spain |   2.371473   .2596354     9.13   0.000     1.862597
>     2.880349
                          Italy |   1.792794   .2468242     7.26   0.000     1.309028
>     2.276561
                         France |   1.767037   .2161379     8.18   0.000     1.343414
>     2.190659
                        Denmark |  -.1891933   .2572554    -0.74   0.462    -.6934047
>     .3150181
                    Switzerland |   .2468488   .2370731     1.04   0.298     -.217806
>     .7115036
                        Belgium |   .5491337   .2070348     2.65   0.008     .1433529
>     .9549144
                 Czech Republic |   1.631587   .2720398     6.00   0.000     1.098399
>     2.164775
                       Slovenia |   1.492036   .3038772     4.91   0.000     .8964481
>     2.087625
-------------------------------+----------------------------------------------------
                        /phi1_1 |          1  (constrained)
                        /phi1_2 |   .8485698   .0263712    32.18   0.000     .7968832
>     .9002563
                        /phi1_3 |   .6254969    .019893    31.44   0.000     .5865073
>     .6644864
                        /phi1_4 |   .3317059    .015479    21.43   0.000     .3013676
>     .3620442
                        /phi1_5 |          0  (base outcome)
-------------------------------+----------------------------------------------------
                        /theta1 |   15.78036   5.846092     2.70   0.007      4.32223
>     27.23849
                        /theta2 |    14.7524   4.989666     2.96   0.003     4.972834
>     24.53197
                        /theta3 |    12.6951   3.699853     3.43   0.001      5.44352
>     19.94668
                        /theta4 |   7.502783   1.988826     3.77   0.000     3.604755
>     11.40081
                        /theta5 |          0  (base outcome)
------------------------------------------------------------------------------
(sphus=Poor is the base outcome)
results_imp1_ste_13_2.xls
dir : seeout

Predictive margins                                      Number of obs = 22,899
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))
```

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]
_predict#rhfo#ghto					
1#None#None	.0468288	.0022322	20.98	0.000	.0424538 .0512037
1#None#At least 1	.0605347	.0036556	16.56	0.000	.0533698 .0676996
1#None#Not applicable	.0486393	.0025614	18.99	0.000	.0436191 .0536595
1#At least 1#None	.0322954	.0026872	12.02	0.000	.0270287 .0375621
1#At least 1#At least 1	.0529687	.0057426	9.22	0.000	.0417133 .064224
1#At least 1#Not applicable	.0333773	.0037415	8.92	0.000	.0260442 .0407105
2#None#None	.1219514	.0038191	31.93	0.000	.1144662 .1294367
2#None#At least 1	.1434947	.0053705	26.72	0.000	.1329686 .1540207
2#None#Not applicable	.124962	.0043747	28.56	0.000	.1163878 .1335363
2#At least 1#None	.0954937	.0050249	19.00	0.000	.0856452 .1053423
2#At least 1#At least 1	.1319466	.0091778	14.38	0.000	.1139583 .1499348
2#At least 1#Not applicable	.0976256	.0072358	13.49	0.000	.0834437 .1118075
3#None#None	.3915018	.0055308	70.79	0.000	.3806617 .402342
3#None#At least 1	.4099275	.0065639	62.45	0.000	.3970625 .4227924
3#None#Not applicable	.3944622	.0058924	66.94	0.000	.3829133 .4060112
3#At least 1#None	.3594356	.0074385	48.32	0.000	.3448563 .3740149
3#At least 1#At least 1	.4008369	.0091392	43.86	0.000	.3829244 .4187495
3#At least 1#Not applicable	.3624557	.0104545	34.67	0.000	.3419654 .3829461
4#None#None	.3158444	.0053959	58.53	0.000	.3052686 .3264202
4#None#At least 1	.2902925	.0068072	42.65	0.000	.2769507 .3036343
4#None#Not applicable	.3123914	.0057382	54.44	0.000	.3011447 .3236382
4#At least 1#None	.3439459	.0073262	46.95	0.000	.3295868 .3583051
4#At least 1#At least 1	.3042226	.0111273	27.34	0.000	.2824135 .3260318
4#At least 1#Not applicable	.3418597	.0087618	39.02	0.000	.3246869 .3590325
5#None#None	.1238736	.0042707	29.01	0.000	.1155032 .132244
5#None#At least 1	.0957507	.0059029	16.22	0.000	.0841812 .1073202
5#None#Not applicable	.119545	.0052017	22.98	0.000	.1093499 .1297401
5#At least 1#None	.1688293	.0081083	20.82	0.000	.1529373 .1847213
5#At least 1#At least 1	.1100252	.0115967	9.49	0.000	.087296 .1327544
5#At least 1#Not applicable	.1646817	.013038	12.63	0.000	.1391276 .1902358

```
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -1.933e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.332e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -69148794  (not concave)
Iteration 3:  Log pseudolikelihood =  -53103821  (not concave)
Iteration 4:  Log pseudolikelihood =  -48641460  (not concave)
Iteration 5:  Log pseudolikelihood =  -48200251  (not concave)
Iteration 6:  Log pseudolikelihood =  -47985669  (not concave)
Iteration 7:  Log pseudolikelihood =  -47839482  (not concave)
Iteration 8:  Log pseudolikelihood =  -47704241
Iteration 9:  Log pseudolikelihood =  -47534267  (not concave)
Iteration 10: Log pseudolikelihood =  -46935253  (not concave)
Iteration 11: Log pseudolikelihood =  -46828850
Iteration 12: Log pseudolikelihood =  -46722861
Iteration 13: Log pseudolikelihood =  -46699000
Iteration 14: Log pseudolikelihood =  -46645649  (not concave)
Iteration 15: Log pseudolikelihood =  -46630273
Iteration 16: Log pseudolikelihood =  -46618068
Iteration 17: Log pseudolikelihood =  -46615384
Iteration 18: Log pseudolikelihood =  -46615168
Iteration 19: Log pseudolikelihood =  -46615167
Iteration 20: Log pseudolikelihood =  -46615167

Stereotype logistic regression                        Number of obs =  18,545
                                                      Wald chi2(32) = 1109.41
Log pseudolikelihood = -46615167                      Prob > chi2   =  0.0000

 ( 1)  [phi1_1]_cons = 1
```

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.6163267	.1710305	3.60	0.000	.281113	.9515403
c.age#c.age	-.0036308	.0011172	-3.25	0.001	-.0058204	-.0014412
yedu	-.051327	.045566	-1.13	0.260	-.1406347	.0379807
c.yedu#c.yedu	-.0018778	.0020421	-0.92	0.358	-.0058803	.0021247
thinc	-.4380313	.1570432	-2.79	0.005	-.7458303	-.1302324
c.thinc#c.thinc	-.0462968	.0710172	-0.65	0.514	-.185488	.0928944
gender						
Female	.0807349	.1382641	0.58	0.559	-.1902578	.3517276
0.covid	0	(omitted)				
household						
Couple, both responding	.3569411	.1567858	2.28	0.023	.0496466	.6642355
Multiple, at least 1 responding	.6101399	.2605525	2.34	0.019	.0994664	1.120813
cjs						
Not applicable	2.812609	.834813	3.37	0.001	1.176406	4.448813
Retired	1.51072	.4579809	3.30	0.001	.6130939	2.408346
Unemployed	.3850121	.85365	0.45	0.652	-1.288111	

```
>     2.058135
   Permanently sick or disabled |   3.398477   .7548498     4.50   0.000     1.918999
>     4.877955
                      Homemaker |   1.718921   .4965776     3.46   0.001     .7456465
>     2.692195
                          Other |   1.593343   .5550091     2.87   0.004     .5055454
>     2.681141
                                |
                           gali |
                        Limited |   5.250076   .2057337    25.52   0.000     4.846845
>     5.653307
                        lifesat |  -1.097217   .1787776    -6.14   0.000    -1.447614
>     -.746819
                                |
             c.lifesat#c.lifesat |    .022956   .0128726     1.78   0.075    -.0022738
>     .0481858
                                |
                           naly |
                      At least 1 |  -.2956032   .1587548    -1.86   0.063    -.6067569
>     .0155506
                  Not applicable |          0  (omitted)
                                |
                           rhfo |
                      At least 1 |   1.289444   .1772777     7.27   0.000     .9419859
>     1.636902
                                |
                           ghto |
                      At least 1 |  -.7269739   .1925923    -3.77   0.000    -1.104448
>    -.3494999
                                |
                       rhfo#ghto |
          At least 1#At least 1 |  -1.080128   .3548236    -3.04   0.002     -1.77557
>    -.3846868
                                |
                        country |
                        Austria |   .5299173   .2157344     2.46   0.014     .1070856
>     .9527491
                        Germany |   1.809045   .2545619     7.11   0.000     1.310113
>     2.307977
                          Spain |   2.220254   .2606475     8.52   0.000     1.709394
>     2.731113
                          Italy |   1.741585   .2440075     7.14   0.000     1.263339
>     2.219831
                         France |   1.631438   .2171961     7.51   0.000     1.205742
>     2.057135
                        Denmark |  -.4893256   .2704909    -1.81   0.070    -1.019478
>     .0408267
                    Switzerland |   .3568654   .2542701     1.40   0.160    -.1414948
>     .8552257
                        Belgium |   .3236053    .209503     1.54   0.122    -.0870129
>     .7342236
                  Czech Republic |   1.849175   .2753267     6.72   0.000     1.309544
>     2.388805
                        Slovenia |   1.932945   .3045086     6.35   0.000     1.336119
>     2.529771
-------------------------------+----------------------------------------------------------
                       /phi1_1 |          1  (constrained)
                       /phi1_2 |   .8812794   .0257981    34.16   0.000     .8307161
>     .9318427
                       /phi1_3 |   .6584668   .0195981    33.60   0.000     .6200552
>     .6968784
                       /phi1_4 |   .3524016   .0165596    21.28   0.000     .3199454
>     .3848578
                       /phi1_5 |          0  (base outcome)
-------------------------------+----------------------------------------------------------
                       /theta1 |   22.96818    6.51932     3.52   0.000     10.19055
>     35.74581
                       /theta2 |   21.65243   5.734084     3.78   0.000     10.41383
>     32.89103
                       /theta3 |   18.01558   4.317853     4.17   0.000     9.552739
```

```
>     26.47841
                        /theta4 |   10.52142   2.375867     4.43   0.000     5.864809
>     15.17804
                        /theta5 |          0  (base outcome)
------------------------------------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_15_2.xls
dir : seeout

Predictive margins                                      Number of obs = 18,545
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------------------------------------------------------
  ____
                             |            Delta-method
                             |     Margin   std. err.      z    P>|z|     [95% conf. inter
> val]
-----------------------------+------------------------------------------------------------
  ____
      _predict#rhfo#ghto     |
            1#None#None      |   .042265   .0024019    17.60   0.000     .0375573    .046
> 9727
      1#None#At least 1      |  .0524561   .0033413    15.70   0.000     .0459071     .05
> 9005
      1#At least 1#None      |  .0278246   .0023817    11.68   0.000     .0231565    .032
> 4927
1#At least 1#At least 1      |  .0493613   .0046781    10.55   0.000     .0401925    .058
> 5302
            2#None#None      |  .1241061   .0040653    30.53   0.000     .1161382    .132
> 0739
      2#None#At least 1      |  .1443165   .0061283    23.55   0.000     .1323053    .156
> 3278
      2#At least 1#None      |  .0918177   .0049313    18.62   0.000     .0821524    .101
> 4829
2#At least 1#At least 1      |   .138367   .0080876    17.11   0.000     .1225156    .154
> 2184
            3#None#None      |  .3876583   .0062603    61.92   0.000     .3753883    .399
> 9283
      3#None#At least 1      |  .4062016   .0079494    51.10   0.000      .390621    .421
> 7821
      3#At least 1#None      |   .346243   .0070177    49.34   0.000     .3324886    .359
> 9975
3#At least 1#At least 1      |  .4012661   .0088638    45.27   0.000     .3838933    .418
> 6389
            4#None#None      |  .3320916   .0058629    56.64   0.000     .3206006    .343
> 5827
      4#None#At least 1      |  .3078379   .0079439    38.75   0.000      .292268    .323
> 4077
      4#At least 1#None      |  .3662366   .0075051    48.80   0.000     .3515269    .380
> 9463
4#At least 1#At least 1      |  .3151331   .0103242    30.52   0.000     .2948981    .335
> 3681
            5#None#None      |   .113879    .004235    26.89   0.000     .1055786    .122
> 1794
      5#None#At least 1      |   .089188   .0063162    14.12   0.000     .0768086    .101
> 5674
      5#At least 1#None      |   .167878   .0074558    22.52   0.000      .153265    .182
> 4911
5#At least 1#At least 1      |  .0958724   .0089995    10.65   0.000     .0782337    .113
> 5111
-----------------------------+------------------------------------------------------------
  ____
slog15_2.xls
dir : seeout
note: 0.covid omitted because of collinearity.
```

```
note: 2.naly omitted because of collinearity.
note: 2.ghto omitted because of collinearity.
note: 0.rhfo#2.ghto identifies no observations in the sample.
note: 1.rhfo#2.ghto identifies no observations in the sample.
note: 2.rhfo#0.ghto identifies no observations in the sample.
note: 2.rhfo#1.ghto identifies no observations in the sample.
note: 2.rhfo#2.ghto omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -1.985e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.086e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -76169458  (not concave)
Iteration 3:  Log pseudolikelihood =  -65241616  (not concave)
Iteration 4:  Log pseudolikelihood =  -51978458  (not concave)
Iteration 5:  Log pseudolikelihood =  -49436160  (not concave)
Iteration 6:  Log pseudolikelihood =  -48428787  (not concave)
Iteration 7:  Log pseudolikelihood =  -47585266  (not concave)
Iteration 8:  Log pseudolikelihood =  -46997121  (not concave)
Iteration 9:  Log pseudolikelihood =  -46219103  (not concave)
Iteration 10: Log pseudolikelihood =  -45603778  (not concave)
Iteration 11: Log pseudolikelihood =  -44652807  (not concave)
Iteration 12: Log pseudolikelihood =  -44141773  (not concave)
Iteration 13: Log pseudolikelihood =  -42975638  (not concave)
Iteration 14: Log pseudolikelihood =  -42555614  (not concave)
Iteration 15: Log pseudolikelihood =  -42444541  (not concave)
Iteration 16: Log pseudolikelihood =  -42319358  (not concave)
Iteration 17: Log pseudolikelihood =  -42175052  (not concave)
Iteration 18: Log pseudolikelihood =  -42124388  (not concave)
Iteration 19: Log pseudolikelihood =  -42085741  (not concave)
Iteration 20: Log pseudolikelihood =  -42051967  (not concave)
Iteration 21: Log pseudolikelihood =  -42019006
Iteration 22: Log pseudolikelihood =  -41930805  (not concave)
Iteration 23: Log pseudolikelihood =  -41849318
Iteration 24: Log pseudolikelihood =  -41821915  (not concave)
Iteration 25: Log pseudolikelihood =  -41767926
Iteration 26: Log pseudolikelihood =  -41755912
Iteration 27: Log pseudolikelihood =  -41751737
Iteration 28: Log pseudolikelihood =  -41750022
Iteration 29: Log pseudolikelihood =  -41749967
Iteration 30: Log pseudolikelihood =  -41749967

Stereotype logistic regression                          Number of obs = 16,181
                                                        Wald chi2(33) = 858.18
Log pseudolikelihood = -41749967                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
------------------------------------------------------------------------------
             |               Robust
       sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]
-------------+----------------------------------------------------------------
         age |   .2671142   .1779883     1.50   0.133    -.0817365
>     .6159649
             |
 c.age#c.age |  -.0013679   .0011488    -1.19   0.234    -.0036195
>     .0008836
             |
        yedu |  -.0315801   .0468229    -0.67   0.500    -.1233512
>      .060191
             |
 c.yedu#c.yedu |  -.0023189   .0021296    -1.09   0.276    -.0064927
>      .001855
             |
       thinc |  -.3105099   .1591946    -1.95   0.051    -.6225256
>     .0015057
             |
 c.thinc#c.thinc |   -.137333   .0873593    -1.57   0.116    -.3085542
>     .0338881
             |
      gender |
     Female  |  -.2167008   .1375281    -1.58   0.115    -.4862509
```

```
>     .0528493
                      0.covid |          0  (omitted)
                              |
                    household |
       Couple, both responding |   .1471686    .149912     0.98   0.326    -.1466536
>     .4409908
Multiple, at least 1 responding |   .1469532   .2741845     0.54   0.592    -.3904384
>     .6843449
                              |
                         cjs  |
                Not applicable |    2.95675   .7085681     4.17   0.000     1.567982
>     4.345518
                       Retired |   1.248651   .5470547     2.28   0.022     .1764438
>     2.320859
                    Unemployed |   .4104038   1.724838     0.24   0.812    -2.970216
>     3.791023
   Permanently sick or disabled |   4.204513   .8351003     5.03   0.000     2.567747
>     5.841279
                     Homemaker |   1.557784   .5797566     2.69   0.007      .421482
>     2.694086
                         Other |   1.033019   .7333381     1.41   0.159    -.4042971
>     2.470335
                              |
                          gali |
                       Limited |   4.825798   .2139021    22.56   0.000     4.406558
>     5.245038
                        lifesat |  -.2688755   .2039877    -1.32   0.187     -.668684
>     .1309331
                              |
             c.lifesat#c.lifesat |  -.0291421   .0145646    -2.00   0.045    -.0576881
>    -.0005961
                              |
                          naly |
                     At least 1 |  -.3465656   .1674011    -2.07   0.038    -.6746657
>    -.0184656
                Not applicable |          0  (omitted)
                              |
                          rhfo |
                     At least 1 |   .8047986   .2316622     3.47   0.001      .350749
>     1.258848
                Not applicable |   .1277986   .2436628     0.52   0.600    -.3497716
>     .6053689
                              |
                          ghto |
                     At least 1 |   -.750505   .2338106    -3.21   0.001    -1.208765
>    -.2922446
                Not applicable |          0  (omitted)
                              |
                       rhfo#ghto |
            None#Not applicable |          0  (empty)
         At least 1#At least 1 |   .7608888   .4269183     1.78   0.075    -.0758557
>     1.597633
      At least 1#Not applicable |          0  (empty)
            Not applicable#None |          0  (empty)
      Not applicable#At least 1 |          0  (empty)
  Not applicable#Not applicable |          0  (omitted)
                              |
                       country |
                       Austria |   .6995504   .2274906     3.08   0.002      .253677
>     1.145424
                       Germany |   1.559658   .2651185     5.88   0.000     1.040036
>     2.079281
                         Spain |   2.112618   .2675741     7.90   0.000     1.588183
>     2.637054
                         Italy |    1.59401   .2503876     6.37   0.000      1.10326
>     2.084761
                        France |   1.088848   .2241581     4.86   0.000     .6495065
>      1.52819
                       Denmark |  -.8033011   .2708192    -2.97   0.003    -1.334097
>    -.2725052
                   Switzerland |   .0939882   .2387022     0.39   0.694    -.3738595
>     .5618359
```

```
                     Belgium |  -.0306502   .2122162    -0.14   0.885    -.4465864
>     .3852859
              Czech Republic |   .2125998   .2399143     0.89   0.376    -.2576236
>     .6828232
                    Slovenia |   1.561623   .2859715     5.46   0.000     1.001129
>     2.122116
------------------------------+----------------------------------------------------
  ____________
                     /phi1_1 |          1  (constrained)
                     /phi1_2 |   .9164664   .0314648    29.13   0.000     .8547964
>     .9781363
                     /phi1_3 |   .6693841   .0228174    29.34   0.000     .6246628
>     .7141054
                     /phi1_4 |   .3478442   .0195474    17.79   0.000      .309532
>     .3861564
                     /phi1_5 |          0  (base outcome)
------------------------------+----------------------------------------------------
  ____________
                     /theta1 |   11.85017   6.917277     1.71   0.087    -1.707444
>     25.40778
                     /theta2 |   12.13803   6.367471     1.91   0.057    -.3419878
>     24.61804
                     /theta3 |    10.6663   4.659584     2.29   0.022      1.53368
>     19.79891
                     /theta4 |   6.358903   2.463094     2.58   0.010     1.531328
>     11.18648
                     /theta5 |          0  (base outcome)
------------------------------+----------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_17_2.xls
dir : seeout

Predictive margins                                      Number of obs = 16,181
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------+----------------------------------------------------
  ____________
                              |            Delta-method
                              |     Margin   std. err.      z    P>|z|     [95% con
> f. interval]
------------------------------+----------------------------------------------------
  ____________
             _predict#rhfo#ghto |
                   1#None#None |   .0398013   .0025768    15.45   0.000      .034751
>     .0448517
             1#None#At least 1 |   .0498381   .0039413    12.65   0.000     .0421134
>     .0575628
         1#None#Not applicable |          .  (not estimable)
             1#At least 1#None |    .030686   .0028449    10.79   0.000     .0251101
>     .0362618
       1#At least 1#At least 1 |   .0305791   .0037305     8.20   0.000     .0232673
>     .0378908
   1#At least 1#Not applicable |          .  (not estimable)
         1#Not applicable#None |          .  (not estimable)
   1#Not applicable#At least 1 |          .  (not estimable)
1#Not applicable#Not applicable |   .0382423   .0029294    13.05   0.000     .0325009
>     .0439838
                   2#None#None |   .1185456   .0050062    23.68   0.000     .1087337
>     .1283575
             2#None#At least 1 |   .1415782   .0080869    17.51   0.000     .1257282
>     .1574282
         2#None#Not applicable |          .  (not estimable)
             2#At least 1#None |   .0961925   .0064506    14.91   0.000     .0835496
>     .1088353
       2#At least 1#At least 1 |   .0959209   .0085296    11.25   0.000     .0792032
```

```
>     .1126385
    2#At least 1#Not applicable |          .  (not estimable)
          2#Not applicable#None |          .  (not estimable)
    2#Not applicable#At least 1 |          .  (not estimable)
2#Not applicable#Not applicable |   .1148278    .006256    18.35   0.000     .1025663
>     .1270893
                     3#None#None |   .3811004   .0073055    52.17   0.000     .3667819
>     .3954189
              3#None#At least 1 |   .4029032   .0094402    42.68   0.000     .3844008
>     .4214056
          3#None#Not applicable |          .  (not estimable)
              3#At least 1#None |   .3530403   .0093553    37.74   0.000     .3347042
>     .3713764
        3#At least 1#At least 1 |    .352651   .0132546    26.61   0.000     .3266725
>     .3786295
    3#At least 1#Not applicable |          .  (not estimable)
          3#Not applicable#None |          .  (not estimable)
    3#Not applicable#At least 1 |          .  (not estimable)
3#Not applicable#Not applicable |   .3769435   .0078053    48.29   0.000     .3616455
>     .3922415
                     4#None#None |   .3342842   .0070728    47.26   0.000     .3204217
>     .3481467
              4#None#At least 1 |   .3080958   .0101906    30.23   0.000     .2881225
>      .328069
          4#None#Not applicable |          .  (not estimable)
              4#At least 1#None |   .3576709   .0085528    41.82   0.000     .3409077
>     .3744341
        4#At least 1#At least 1 |   .3579368   .0103675    34.52   0.000     .3376169
>     .3782566
    4#At least 1#Not applicable |          .  (not estimable)
          4#Not applicable#None |          .  (not estimable)
    4#Not applicable#At least 1 |          .  (not estimable)
4#Not applicable#Not applicable |   .3383479   .0077179    43.84   0.000      .323221
>     .3534748
                     5#None#None |   .1262685   .0059307    21.29   0.000     .1146445
>     .1378925
              5#None#At least 1 |   .0975847   .0085129    11.46   0.000     .0808997
>     .1142698
          5#None#Not applicable |          .  (not estimable)
              5#At least 1#None |   .1624103   .0105686    15.37   0.000     .1416962
>     .1831245
        5#At least 1#At least 1 |   .1629123   .0156842    10.39   0.000     .1321717
>     .1936528
    5#At least 1#Not applicable |          .  (not estimable)
          5#Not applicable#None |          .  (not estimable)
    5#Not applicable#At least 1 |          .  (not estimable)
5#Not applicable#Not applicable |   .1316384   .0080354    16.38   0.000     .1158892
>     .1473876
------------------------------------------------------------------------------------------
```

slog17_2.xls
dir : seeout
note: **0.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -2.075e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.408e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -90836834  (not concave)
Iteration 3:  Log pseudolikelihood =  -76133189  (not concave)
Iteration 4:  Log pseudolikelihood =  -45197626  (not concave)
Iteration 5:  Log pseudolikelihood =  -36701513  (not concave)
Iteration 6:  Log pseudolikelihood =  -32569706  (not concave)
Iteration 7:  Log pseudolikelihood =  -30294266  (not concave)
Iteration 8:  Log pseudolikelihood =  -29716093  (not concave)
Iteration 9:  Log pseudolikelihood =  -29578352  (not concave)
Iteration 10: Log pseudolikelihood =  -29337864  (not concave)
Iteration 11: Log pseudolikelihood =  -29186956  (not concave)
Iteration 12: Log pseudolikelihood =  -29117789  (not concave)
Iteration 13: Log pseudolikelihood =  -29082347  (not concave)
Iteration 14: Log pseudolikelihood =  -29048297  (not concave)
Iteration 15: Log pseudolikelihood =  -29024424  (not concave)
Iteration 16: Log pseudolikelihood =  -29001853  (not concave)
```

```
Iteration 17: Log pseudolikelihood =  -28964425  (not concave)
Iteration 18: Log pseudolikelihood =  -28932988  (not concave)
Iteration 19: Log pseudolikelihood =  -28885432  (not concave)
Iteration 20: Log pseudolikelihood =  -28841576  (not concave)
Iteration 21: Log pseudolikelihood =  -28800582  (not concave)
Iteration 22: Log pseudolikelihood =  -28775756  (not concave)
Iteration 23: Log pseudolikelihood =  -28765358  (not concave)
Iteration 24: Log pseudolikelihood =  -28755356  (not concave)
Iteration 25: Log pseudolikelihood =  -28745808  (not concave)
Iteration 26: Log pseudolikelihood =  -28736505  (not concave)
Iteration 27: Log pseudolikelihood =  -28727447  (not concave)
Iteration 28: Log pseudolikelihood =  -28718663  (not concave)
Iteration 29: Log pseudolikelihood =  -28710120  (not concave)
Iteration 30: Log pseudolikelihood =  -28701785  (not concave)
Iteration 31: Log pseudolikelihood =  -28693654  (not concave)
Iteration 32: Log pseudolikelihood =  -28685715  (not concave)
Iteration 33: Log pseudolikelihood =  -28677951  (not concave)
Iteration 34: Log pseudolikelihood =  -28670348  (not concave)
Iteration 35: Log pseudolikelihood =  -28662897  (not concave)
Iteration 36: Log pseudolikelihood =  -28655584  (not concave)
Iteration 37: Log pseudolikelihood =  -28648402  (not concave)
Iteration 38: Log pseudolikelihood =  -28641338  (not concave)
Iteration 39: Log pseudolikelihood =  -28634386  (not concave)
Iteration 40: Log pseudolikelihood =  -28627534  (not concave)
Iteration 41: Log pseudolikelihood =  -28620778  (not concave)
Iteration 42: Log pseudolikelihood =  -28614107  (not concave)
Iteration 43: Log pseudolikelihood =  -28607518  (not concave)
Iteration 44: Log pseudolikelihood =  -28601001  (not concave)
Iteration 45: Log pseudolikelihood =  -28594554  (not concave)
Iteration 46: Log pseudolikelihood =  -28588167  (not concave)
Iteration 47: Log pseudolikelihood =  -28581839  (not concave)
Iteration 48: Log pseudolikelihood =  -28575563  (not concave)
Iteration 49: Log pseudolikelihood =  -28569337  (not concave)
Iteration 50: Log pseudolikelihood =  -28563153  (not concave)
Iteration 51: Log pseudolikelihood =  -28557012  (not concave)
Iteration 52: Log pseudolikelihood =  -28550906  (not concave)
Iteration 53: Log pseudolikelihood =  -28544838  (not concave)
Iteration 54: Log pseudolikelihood =  -28538798  (not concave)
Iteration 55: Log pseudolikelihood =  -28532790  (not concave)
Iteration 56: Log pseudolikelihood =  -28526806  (not concave)
Iteration 57: Log pseudolikelihood =  -28520848  (not concave)
Iteration 58: Log pseudolikelihood =  -28514911  (not concave)
Iteration 59: Log pseudolikelihood =  -28508997  (not concave)
Iteration 60: Log pseudolikelihood =  -28503099  (not concave)
Iteration 61: Log pseudolikelihood =  -28497222  (not concave)
Iteration 62: Log pseudolikelihood =  -28491358  (not concave)
Iteration 63: Log pseudolikelihood =  -28485512  (not concave)
Iteration 64: Log pseudolikelihood =  -28479677  (not concave)
Iteration 65: Log pseudolikelihood =  -28473858  (not concave)
Iteration 66: Log pseudolikelihood =  -28468048  (not concave)
Iteration 67: Log pseudolikelihood =  -28462253  (not concave)
Iteration 68: Log pseudolikelihood =  -28456465  (not concave)
Iteration 69: Log pseudolikelihood =  -28450691  (not concave)
Iteration 70: Log pseudolikelihood =  -28444924  (not concave)
Iteration 71: Log pseudolikelihood =  -28439169  (not concave)
Iteration 72: Log pseudolikelihood =  -28433420  (not concave)
Iteration 73: Log pseudolikelihood =  -28427684  (not concave)
Iteration 74: Log pseudolikelihood =  -28421953  (not concave)
Iteration 75: Log pseudolikelihood =  -28416235  (not concave)
Iteration 76: Log pseudolikelihood =  -28410522  (not concave)
Iteration 77: Log pseudolikelihood =  -28404821  (not concave)
Iteration 78: Log pseudolikelihood =  -28399125  (not concave)
Iteration 79: Log pseudolikelihood =  -28393442  (not concave)
Iteration 80: Log pseudolikelihood =  -28387765  (not concave)
Iteration 81: Log pseudolikelihood =  -28382101  (not concave)
Iteration 82: Log pseudolikelihood =  -28376444  (not concave)
Iteration 83: Log pseudolikelihood =  -28370799  (not concave)
Iteration 84: Log pseudolikelihood =  -28365162  (not concave)
Iteration 85: Log pseudolikelihood =  -28359539  (not concave)
Iteration 86: Log pseudolikelihood =  -28353924  (not concave)
Iteration 87: Log pseudolikelihood =  -28348325  (not concave)
Iteration 88: Log pseudolikelihood =  -28342734  (not concave)
```

```
Iteration 89: Log pseudolikelihood =  -28337159  (not concave)
Iteration 90: Log pseudolikelihood =  -28331594  (not concave)
Iteration 91: Log pseudolikelihood =  -28326047  (not concave)
Iteration 92: Log pseudolikelihood =  -28320510  (not concave)
Iteration 93: Log pseudolikelihood =  -28314992  (not concave)
Iteration 94: Log pseudolikelihood =  -28309487  (not concave)
Iteration 95: Log pseudolikelihood =  -28304001  (not concave)
Iteration 96: Log pseudolikelihood =  -28298529  (not concave)
Iteration 97: Log pseudolikelihood =  -28293078  (not concave)
Iteration 98: Log pseudolikelihood =  -28287643  (not concave)
Iteration 99: Log pseudolikelihood =  -28282229  (not concave)
Iteration 100: Log pseudolikelihood =  -28276833  (not concave)
Iteration 101: Log pseudolikelihood =  -28271460  (not concave)
Iteration 102: Log pseudolikelihood =  -28266105  (not concave)
Iteration 103: Log pseudolikelihood =  -28260776  (not concave)
Iteration 104: Log pseudolikelihood =  -28255467  (not concave)
Iteration 105: Log pseudolikelihood =  -28250184  (not concave)
Iteration 106: Log pseudolikelihood =  -28244923  (not concave)
Iteration 107: Log pseudolikelihood =  -28239690  (not concave)
Iteration 108: Log pseudolikelihood =  -28234480  (not concave)
Iteration 109: Log pseudolikelihood =  -28229299  (not concave)
Iteration 110: Log pseudolikelihood =  -28224144  (not concave)
Iteration 111: Log pseudolikelihood =  -28219019  (not concave)
Iteration 112: Log pseudolikelihood =  -28213921  (not concave)
Iteration 113: Log pseudolikelihood =  -28208855  (not concave)
Iteration 114: Log pseudolikelihood =  -28203817  (not concave)
Iteration 115: Log pseudolikelihood =  -28198814  (not concave)
Iteration 116: Log pseudolikelihood =  -28193839  (not concave)
Iteration 117: Log pseudolikelihood =  -28188900  (not concave)
Iteration 118: Log pseudolikelihood =  -28183992  (not concave)
Iteration 119: Log pseudolikelihood =  -28179121  (not concave)
Iteration 120: Log pseudolikelihood =  -28174282  (not concave)
Iteration 121: Log pseudolikelihood =  -28169482  (not concave)
Iteration 122: Log pseudolikelihood =  -28164715  (not concave)
Iteration 123: Log pseudolikelihood =  -28159988
Iteration 124: Log pseudolikelihood =  -27968502  (not concave)
Iteration 125: Log pseudolikelihood =  -27913491  (not concave)
Iteration 126: Log pseudolikelihood =  -27904034
Iteration 127: Log pseudolikelihood =  -27868197
Iteration 128: Log pseudolikelihood =  -27858682  (not concave)
Iteration 129: Log pseudolikelihood =  -27853877
Iteration 130: Log pseudolikelihood =  -27852424
Iteration 131: Log pseudolikelihood =  -27852357
Iteration 132: Log pseudolikelihood =  -27852356
Iteration 133: Log pseudolikelihood =  -27852356

Stereotype logistic regression                          Number of obs = 10,710
                                                        Wald chi2(33) = 747.97
Log pseudolikelihood = -27852356                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
------------------------------------------------------------------------------
             |               Robust
       sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]
-------------+----------------------------------------------------------------
         age |   .5691033   .2598316     2.19   0.029     .0598426
>     1.078364
             |
 c.age#c.age |  -.0033379   .0016252    -2.05   0.040    -.0065233
>    -.0001525
             |
        yedu |  -.0336905   .0614962    -0.55   0.584    -.1542208
>     .0868399
             |
 c.yedu#c.yedu |  -.0021938    .002745    -0.80   0.424    -.0075739
>     .0031864
             |
       thinc |  -.1817738   .1857328    -0.98   0.328    -.5458034
>     .1822558
```

```
                                 |
                c.thinc#c.thinc  |  -.0439237   .0882649    -0.50   0.619    -.2169197
>     .1290723
                                 |
                         gender  |
                         Female  |  -.0275767   .1806742    -0.15   0.879    -.3816916
>     .3265383
                        0.covid  |          0  (omitted)
                                 |
                      household  |
        Couple, both responding  |   .5335276   .2075748     2.57   0.010     .1266885
>     .9403668
Multiple, at least 1 responding  |   .5224621   .3483951     1.50   0.134    -.1603798
>     1.205304
                                 |
                            cjs  |
                Not applicable   |   3.741325   1.253356     2.99   0.003     1.284793
>     6.197858
                       Retired   |   3.763544   1.064542     3.54   0.000      1.67708
>     5.850009
                    Unemployed   |   2.839231    2.58122     1.10   0.271    -2.219867
>      7.89833
   Permanently sick or disabled  |   5.329192   1.328403     4.01   0.000      2.72557
>     7.932814
                     Homemaker   |   3.421781    1.09341     3.13   0.002     1.278737
>     5.564826
                         Other   |   3.609604   1.187404     3.04   0.002     1.282334
>     5.936874
                                 |
                          gali   |
                       Limited   |   5.371324   .2950284    18.21   0.000     4.793079
>     5.949569
                        lifesat  |  -1.365952   .3147278    -4.34   0.000    -1.982808
>    -.7490972
                                 |
            c.lifesat#c.lifesat  |   .0383423   .0220163     1.74   0.082    -.0048089
>     .0814935
                                 |
                          naly   |
                    At least 1   |  -1.043277   .2661001    -3.92   0.000    -1.564824
>    -.5217308
                Not applicable   |          0  (omitted)
                                 |
                          rhfo   |
                    At least 1   |    .773298   .2305595     3.35   0.001     .3214097
>     1.225186
                                 |
                          ghto   |
                    At least 1   |  -.6087192    .242703    -2.51   0.012    -1.084408
>      -.13303
                                 |
                      rhfo#ghto  |
         At least 1#At least 1   |  -.1826237   .4686085    -0.39   0.697    -1.101079
>      .735832
                                 |
                       country   |
                       Austria   |   .4656708   .3039084     1.53   0.125    -.1299788
>      1.06132
                       Germany   |   1.127691   .3416538     3.30   0.001     .4580617
>      1.79732
                   Netherlands   |  -.1862605   .3249801    -0.57   0.567    -.8232097
>     .4506888
                         Spain   |   2.081966   .3656821     5.69   0.000     1.365242
>      2.79869
                         Italy   |   1.293031   .3654177     3.54   0.000     .5768254
>     2.009236
                        France   |   .9548582    .289544     3.30   0.001     .3873624
>     1.522354
                       Denmark   |  -.4582813   .3524208    -1.30   0.193    -1.149013
>     .2324507
                   Switzerland   |  -.1186266   .3350825    -0.35   0.723    -.7753761
>      .538123
```

```
                     Belgium |   .0407666   .3016751     0.14   0.893    -.5505056
>     .6320389
             Czech Republic |  -.2432412    .345636    -0.70   0.482    -.9206753
>     .4341928
                   Slovenia |    1.17262   .3792255     3.09   0.002     .4293513
>     1.915888
------------------------------+----------------------------------------------------
  ____________
                    /phi1_1 |          1  (constrained)
                    /phi1_2 |   .8426237   .0375604    22.43   0.000     .7690066
>     .9162408
                    /phi1_3 |   .6465043   .0282085    22.92   0.000     .5912167
>     .7017919
                    /phi1_4 |   .3397056   .0234205    14.50   0.000     .2938024
>     .3856089
                    /phi1_5 |          0  (base outcome)
------------------------------+----------------------------------------------------
  ____________
                    /theta1 |   21.26436   10.36057     2.05   0.040     .9580181
>     41.57069
                    /theta2 |   19.59884   8.868075     2.21   0.027     2.217729
>     36.97994
                    /theta3 |   16.76695   6.770932     2.48   0.013     3.496166
>     30.03773
                    /theta4 |   9.740128   3.640742     2.68   0.007     2.604405
>     16.87585
                    /theta5 |          0  (base outcome)
------------------------------+----------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_20_2.xls
dir : seeout

Predictive margins                                      Number of obs = 10,710
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------+----------------------------------------------------
  ____
                              |            Delta-method
                              |     Margin   std. err.      z    P>|z|     [95% conf. inter
> val]
------------------------------+----------------------------------------------------
  ____
      _predict#rhfo#ghto      |
            1#None#None       |   .0362429   .0026847    13.50   0.000      .030981    .041
> 5048
      1#None#At least 1       |   .0439398   .0038419    11.44   0.000     .0364098    .051
> 4697
      1#At least 1#None       |   .0279971   .0028359     9.87   0.000     .0224389    .033
> 5553
1#At least 1#At least 1       |   .0364552    .004961     7.35   0.000     .0267319    .046
> 1785
            2#None#None       |   .1291348   .0058248    22.17   0.000     .1177183    .140
> 5512
      2#None#At least 1       |   .1456289   .0078969    18.44   0.000     .1301512    .161
> 1066
      2#At least 1#None       |   .1094392   .0065109    16.81   0.000     .0966781    .122
> 2004
2#At least 1#At least 1       |    .129612    .010815    11.98   0.000     .1084149     .15
> 0809
            3#None#None       |   .3875201   .0081491    47.55   0.000     .3715482    .403
> 4921
      3#None#At least 1       |     .40484    .010471    38.66   0.000     .3843172    .425
> 3628
      3#At least 1#None       |   .3621635   .0095998    37.73   0.000     .3433483    .380
> 9787
```

```
3#At least 1#At least 1 |   .3880696   .0137297    28.26   0.000     .3611598    .414
> 9793
            4#None#None |   .3423638   .0078463    43.63   0.000     .3269853    .357
> 7423
      4#None#At least 1 |    .320329    .010719    29.88   0.000     .2993201    .341
> 3379
      4#At least 1#None |   .3667016   .0098353    37.28   0.000     .3474247    .385
> 9785
4#At least 1#At least 1 |   .3417439   .0142535    23.98   0.000     .3138074    .369
> 6803
            5#None#None |   .1047384   .0054273    19.30   0.000      .094101    .115
> 3757
      5#None#At least 1 |   .0852623   .0074973    11.37   0.000     .0705679    .099
> 9567
      5#At least 1#None |   .1336985   .0085349    15.66   0.000     .1169704    .150
> 4267
5#At least 1#At least 1 |   .1041194   .0135353     7.69   0.000     .0775906    .130
> 6481
------------------------+----------------------------------------------------------------
```

slog20_2.xls
dir : seeout
note: **1.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -1.952e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.599e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.068e+08  (not concave)
Iteration 3:  Log pseudolikelihood =  -77224760  (not concave)
Iteration 4:  Log pseudolikelihood =  -56002940  (not concave)
Iteration 5:  Log pseudolikelihood =  -43452434  (not concave)
Iteration 6:  Log pseudolikelihood =  -40608427  (not concave)
Iteration 7:  Log pseudolikelihood =  -33272769  (not concave)
Iteration 8:  Log pseudolikelihood =  -31776280  (not concave)
Iteration 9:  Log pseudolikelihood =  -30881101  (not concave)
Iteration 10: Log pseudolikelihood =  -30180664  (not concave)
Iteration 11: Log pseudolikelihood =  -29855271  (not concave)
Iteration 12: Log pseudolikelihood =  -29714721  (not concave)
Iteration 13: Log pseudolikelihood =  -29595367  (not concave)
Iteration 14: Log pseudolikelihood =  -29525749  (not concave)
Iteration 15: Log pseudolikelihood =  -29467991  (not concave)
Iteration 16: Log pseudolikelihood =  -29414175  (not concave)
Iteration 17: Log pseudolikelihood =  -29372495  (not concave)
Iteration 18: Log pseudolikelihood =  -29334415  (not concave)
Iteration 19: Log pseudolikelihood =  -29297758  (not concave)
Iteration 20: Log pseudolikelihood =  -29263022  (not concave)
Iteration 21: Log pseudolikelihood =  -29229740  (not concave)
Iteration 22: Log pseudolikelihood =  -29197192  (not concave)
Iteration 23: Log pseudolikelihood =  -29165223  (not concave)
Iteration 24: Log pseudolikelihood =  -29133871  (not concave)
Iteration 25: Log pseudolikelihood =  -29102962  (not concave)
Iteration 26: Log pseudolikelihood =  -29072386  (not concave)
Iteration 27: Log pseudolikelihood =  -29042044  (not concave)
Iteration 28: Log pseudolikelihood =  -29011983  (not concave)
Iteration 29: Log pseudolikelihood =  -28982113  (not concave)
Iteration 30: Log pseudolikelihood =  -28952475  (not concave)
Iteration 31: Log pseudolikelihood =  -28922996  (not concave)
Iteration 32: Log pseudolikelihood =  -28893766  (not concave)
Iteration 33: Log pseudolikelihood =  -28864722  (not concave)
Iteration 34: Log pseudolikelihood =  -28835965  (not concave)
Iteration 35: Log pseudolikelihood =  -28807437  (not concave)
Iteration 36: Log pseudolikelihood =  -28779258  (not concave)
Iteration 37: Log pseudolikelihood =  -28751376  (not concave)
Iteration 38: Log pseudolikelihood =  -28723917  (not concave)
Iteration 39: Log pseudolikelihood =  -28696834  (not concave)
Iteration 40: Log pseudolikelihood =  -28670256  (not concave)
Iteration 41: Log pseudolikelihood =  -28644138  (not concave)
Iteration 42: Log pseudolikelihood =  -28618608  (not concave)
Iteration 43: Log pseudolikelihood =  -28593620  (not concave)
Iteration 44: Log pseudolikelihood =  -28569295  (not concave)
Iteration 45: Log pseudolikelihood =  -28545590  (not concave)
Iteration 46: Log pseudolikelihood =  -28522613  (not concave)
```

```
Iteration 47: Log pseudolikelihood =  -28500319  (not concave)
Iteration 48: Log pseudolikelihood =  -28478804  (not concave)
Iteration 49: Log pseudolikelihood =  -28458019  (not concave)
Iteration 50: Log pseudolikelihood =  -28438044  (not concave)
Iteration 51: Log pseudolikelihood =  -28418830  (not concave)
Iteration 52: Log pseudolikelihood =  -28400439  (not concave)
Iteration 53: Log pseudolikelihood =  -28382818
Iteration 54: Log pseudolikelihood =  -28341992  (not concave)
Iteration 55: Log pseudolikelihood =  -28275098
Iteration 56: Log pseudolikelihood =  -28252878  (not concave)
Iteration 57: Log pseudolikelihood =  -28206729
Iteration 58: Log pseudolikelihood =  -28154220  (not concave)
Iteration 59: Log pseudolikelihood =  -28129374
Iteration 60: Log pseudolikelihood =  -28106226  (not concave)
Iteration 61: Log pseudolikelihood =  -28094460
Iteration 62: Log pseudolikelihood =  -28085915
Iteration 63: Log pseudolikelihood =  -28083772
Iteration 64: Log pseudolikelihood =  -28083573
Iteration 65: Log pseudolikelihood =  -28083572
Iteration 66: Log pseudolikelihood =  -28083572

Stereotype logistic regression                          Number of obs = 11,204
                                                        Wald chi2(32) = 589.53
Log pseudolikelihood = -28083572                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
```

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.8776758	.3149296	2.79	0.005	.2604252	1.494927
c.age#c.age	-.0050887	.0019468	-2.61	0.009	-.0089044	-.001273
yedu	.0428722	.065299	0.66	0.511	-.0851116	.1708559
c.yedu#c.yedu	-.0048284	.0028901	-1.67	0.095	-.0104929	.0008362
thinc	-.2529764	.1941972	-1.30	0.193	-.6335959	.127643
c.thinc#c.thinc	-.0031131	.095453	-0.03	0.974	-.1901976	.1839714
gender						
Female	.1213663	.181737	0.67	0.504	-.2348318	.4775644
1.covid	0	(omitted)				
household						
Couple, both responding	.6974696	.20733	3.36	0.001	.2911103	1.103829
Multiple, at least 1 responding	.3260402	.4019723	0.81	0.417	-.4618109	1.113891
cjs						
Not applicable	-.5501327	1.962434	-0.28	0.779	-4.396433	3.296168
Retired	-.2184802	1.83428	-0.12	0.905	-3.813602	3.376642
Permanently sick or disabled	.9202026	2.097806	0.44	0.661	-3.191421	5.031826
Homemaker	-.6736742	1.856387	-0.36	0.717	-4.312126	2.964778
Other	.3700713	1.945614	0.19	0.849	-3.443262	

```
>     4.183404
                                  |
                         gali     |
                      Limited     |   5.663679   .3047111    18.59   0.000     5.066457
>     6.260902
                      lifesat     |  -.9793724   .2993701    -3.27   0.001    -1.566127
>    -.3926178
                                  |
             c.lifesat#c.lifesat  |   .0004399   .0210031     0.02   0.983    -.0407255
>     .0416052
                                  |
                         naly     |
                    At least 1    |  -.9262754   .2267118    -4.09   0.000    -1.370622
>    -.4819285
                Not applicable    |          0  (omitted)
                                  |
                         rhfo     |
                    At least 1    |   .8170351   .2220488     3.68   0.000     .3818274
>     1.252243
                                  |
                         ghto     |
                    At least 1    |  -.6403601   .2825927    -2.27   0.023    -1.194232
>    -.0864886
                                  |
                      rhfo#ghto   |
         At least 1#At least 1    |    .129604   .5076628     0.26   0.798    -.8653968
>     1.124605
                                  |
                       country    |
                      Austria     |    .996458   .3060834     3.26   0.001     .3965457
>      1.59637
                      Germany     |   1.638167   .3667199     4.47   0.000     .9194091
>     2.356925
                  Netherlands     |   -.119036   .3423242    -0.35   0.728    -.7899791
>      .551907
                        Spain     |     2.3195   .3801101     6.10   0.000     1.574498
>     3.064502
                        Italy     |   1.730199   .3559692     4.86   0.000     1.032512
>     2.427886
                       France     |   1.100332   .3067019     3.59   0.000     .4992075
>     1.701457
                      Denmark     |  -.0589825   .3656578    -0.16   0.872    -.7756587
>     .6576936
                  Switzerland     |   .1794312   .3735364     0.48   0.631    -.5526867
>     .9115491
                      Belgium     |   .2966003   .2991876     0.99   0.322    -.2897967
>     .8829973
               Czech Republic     |   .1185865   .3772671     0.31   0.753    -.6208434
>     .8580163
                     Slovenia     |   1.584125   .3605053     4.39   0.000     .8775481
>     2.290703
-------------------------------------------------------------------------------------------
                       /phi1_1    |          1  (constrained)
                       /phi1_2    |   .8372624   .0370154    22.62   0.000     .7647136
>     .9098112
                       /phi1_3    |    .633158   .0293421    21.58   0.000     .5756487
>     .6906674
                       /phi1_4    |   .3528528   .0240483    14.67   0.000      .305719
>     .3999865
                       /phi1_5    |          0  (base outcome)
-------------------------------------------------------------------------------------------
                       /theta1    |   32.58543    12.8376     2.54   0.011     7.424196
>     57.74667
                       /theta2    |   28.87896   10.67311     2.71   0.007     7.960052
>     49.79787
                       /theta3    |   23.71143   8.088827     2.93   0.003     7.857617
>     39.56523
                       /theta4    |   14.05856   4.507595     3.12   0.002     5.223841
>     22.89329
                       /theta5    |          0  (base outcome)
```

(sphus=Poor is the base outcome)
results_imp1_ste_22_2.xls
dir : seeout

Predictive margins Number of obs = **11,204**
Model VCE: **Robust**

1._predict: **Pr(sphus==Excellent), predict(pr outcome(#1))**
2._predict: **Pr(sphus==Very good), predict(pr outcome(#2))**
3._predict: **Pr(sphus==Good), predict(pr outcome(#3))**
4._predict: **Pr(sphus==Fair), predict(pr outcome(#4))**
5._predict: **Pr(sphus==Poor), predict(pr outcome(#5))**

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]	
_predict#rhfo#ghto						
1#None#None	.0367857	.0026723	13.77	0.000	.0315481	.0420233
1#None#At least 1	.0451922	.00475	9.51	0.000	.0358824	.054502
1#At least 1#None	.0278634	.0028185	9.89	0.000	.0223392	.0333876
1#At least 1#At least 1	.033215	.005013	6.63	0.000	.0233897	.0430404
2#None#None	.1151681	.0049051	23.48	0.000	.1055543	.124782
2#None#At least 1	.130652	.0078678	16.61	0.000	.1152313	.1460727
2#At least 1#None	.0967006	.0056155	17.22	0.000	.0856945	.1077067
2#At least 1#At least 1	.108064	.009563	11.30	0.000	.0893209	.1268071
3#None#None	.3917873	.0079699	49.16	0.000	.3761665	.407408
3#None#At least 1	.409479	.0107906	37.95	0.000	.3883299	.436281
3#At least 1#None	.3656695	.0090979	40.19	0.000	.347838	.385009
3#At least 1#At least 1	.3824204	.0139747	27.37	0.000	.3550306	.408103
4#None#None	.3415579	.0074201	46.03	0.000	.3270148	.356101
4#None#At least 1	.321995	.0109161	29.50	0.000	.3005997	.343902
4#At least 1#None	.3621674	.0090239	40.13	0.000	.344481	.378539
4#At least 1#At least 1	.349899	.012413	28.19	0.000	.32557	.374228
5#None#None	.114701	.0054279	21.13	0.000	.1040625	.123394
5#None#At least 1	.0926819	.0092688	10.00	0.000	.0745154	.108483
5#At least 1#None	.1475991	.008528	17.31	0.000	.1308846	.163137
5#At least 1#At least 1	.1264015	.0155303	8.14	0.000	.0959626	.158405

slog22_2.xls
dir : seeout
name: **The_value_of_a_helping_hand_sup**
log: **C:\Users\hansg\Documents\working\SHARE data\weighting\output_sup_2.smcl**
log type: **smcl**
closed on: **5 Apr 2026, 13:42:29**